

O'NEILL, NE, USA

WMO: 725566

Lat: 42.470N

Lon: 98.688W

Elev: 2031

StdP: 13.65

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-0.7	-13.5	2.9	-5.3	-8.4	3.8	0.0	36.4	18.8	33.2	21.8	10.2	320	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	94.6	70.9	90.8	70.9	88.0	69.8	75.5	86.4	74.0	84.7	72.4	83.2	13.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	129.7	81.2	70.6	121.5	79.9	69.2	115.6	78.9	40.5	86.5	38.9	85.3	37.5	82.3	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.0	26.3	23.5	-13.7	99.6	4.5	4.4	-17.0	102.8	-19.6	105.3	-22.2	107.8	-25.5	111.0
			-14.2	78.2	4.3	1.7	-17.3	79.4	-19.8	80.4	-22.2	81.4	-25.3	82.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	23.9	26.8	37.0	47.8	58.4	68.8	75.0	72.2	64.2	50.1	37.4	25.8
	DBStd	20.58	12.96	12.57	12.64	9.92	8.70	7.21	6.11	5.83	8.68	9.74	11.04	11.81
	HDD50	3337	809	651	430	153	23	1	0	0	5	118	396	751
	HDD65	6663	1274	1069	868	520	241	43	5	10	120	470	827	1216
	CDD50	2999	0	2	28	88	283	565	777	687	430	121	18	0
	CDD65	852	0	0	1	5	37	157	317	233	94	8	0	0
	CDH74	8824	0	0	24	97	443	1565	3362	2177	1024	123	9	0
	CDH80	3510	0	0	2	22	124	550	1515	880	384	32	1	0
	WSAvg	10.6	11.1	11.2	11.6	12.3	11.2	9.7	8.9	8.7	10.0	10.9	10.7	10.8
Precipitation	PrecAvg	25.2	0.6	0.6	1.5	2.9	3.9	3.9	2.6	2.8	2.2	2.4	1.0	0.7
	PrecMax	37.2	1.5	1.5	3.1	6.0	8.0	8.9	5.0	6.2	4.9	6.0	2.4	2.8
	PrecMin	18.8	0.1	0.2	0.1	0.9	0.2	1.5	0.7	0.4	0.6	0.0	0.0	0.1
	PrecStd	5.1	0.4	0.4	0.9	1.4	1.9	1.9	1.2	1.8	1.3	1.7	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	65.5	77.4	82.4	89.5	94.4	101.8	98.9	94.3	84.1	73.8	58.7
		MCWB	44.1	47.6	54.5	59.1	65.5	71.6	70.2	70.3	66.8	61.0	52.3	44.2
	2%	DB	51.1	55.5	69.9	75.5	83.1	90.0	95.8	92.0	89.5	77.0	65.8	52.2
		MCWB	40.6	43.1	52.0	56.1	64.5	71.1	71.9	70.7	67.7	58.2	50.0	42.3
	5%	DB	45.2	49.4	62.8	70.3	79.1	86.2	91.3	88.5	84.3	71.9	59.5	45.5
		MCWB	37.2	40.0	50.0	54.3	62.4	69.3	72.3	70.2	66.4	56.5	47.2	37.5
	10%	DB	40.6	43.7	55.7	64.8	74.5	82.4	88.4	84.7	80.5	66.2	54.0	40.9
		MCWB	35.1	37.1	45.6	51.3	60.6	67.3	71.5	69.7	64.4	54.0	44.4	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	49.5	59.3	62.3	70.0	75.9	78.3	76.6	73.5	65.2	55.7	47.9
		MCDB	58.7	63.0	67.0	76.0	82.6	87.1	89.3	87.0	84.5	77.9	66.2	56.7
	2%	WB	41.0	44.4	54.5	58.6	67.0	73.3	76.3	74.5	71.0	61.4	51.4	41.6
		MCDB	50.1	54.3	65.4	71.3	78.8	85.1	87.4	84.7	81.7	70.8	63.4	51.2
	5%	WB	37.7	40.6	50.4	55.5	64.3	71.5	74.8	73.0	69.0	58.4	47.9	38.3
		MCDB	44.2	48.9	62.7	68.1	75.6	82.9	85.8	82.8	79.9	67.7	59.4	45.0
	10%	WB	35.3	37.3	46.1	52.6	61.8	69.4	73.2	71.3	67.0	55.1	44.2	35.2
		MCDB	40.2	43.2	56.0	63.3	72.4	80.4	84.1	81.3	77.3	65.5	53.4	40.4
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.2	21.2	21.6	20.8	21.0	22.1	21.8	23.5	22.3	21.2	17.8
		MCDBR	25.3	28.2	32.5	31.5	27.2	24.9	26.4	27.0	28.8	30.7	31.0	24.9
		MCWBR	16.6	17.3	17.6	15.8	13.1	10.9	9.8	10.6	11.8	15.5	17.4	16.3
	5% WB	MCDBR	24.5	26.7	30.2	27.1	23.5	21.9	22.3	21.7	23.7	25.3	29.9	23.8
		MCWBR	16.6	17.0	17.3	14.8	12.5	11.1	10.6	10.3	11.4	14.7	17.3	16.1
Clear-Sky Solar Irradiance	taub		0.265	0.278	0.304	0.342	0.365	0.379	0.392	0.394	0.353	0.321	0.277	0.265
	taud		2.493	2.482	2.456	2.359	2.381	2.379	2.368	2.344	2.420	2.486	2.529	2.485
	Ebn at Noon		285	297	300	294	288	283	278	274	279	276	277	273
	Edh at Noon		23	27	31	37	37	38	38	37	32	27	22	21
All-Sky Solar Radiation	RadAvg		637	932	1239	1580	1865	2119	2154	1825	1475	1024	711	535
	RadStd		56	64	112	92	146	143	86	128	104	115	58	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.19	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page